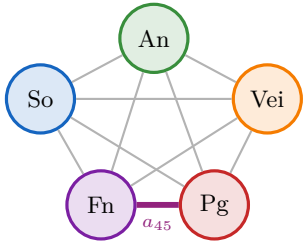
(a) Variational multiphase model



state at a material point:
volume fractions ϕ_i, viabilities ψ_i

$\dot{\Psi} + \Delta + C \rightarrow$ stat

$\Psi = -\frac{1}{2} c^* \, \bar{\boldsymbol{\phi}}^{\mathsf{T}} \mathbf{A} \, \bar{\boldsymbol{\phi}}, \;\; \bar{\phi}_i = \phi_i \psi_i$

$\mathbf{A} = \mathbf{A}^{\mathsf{T}}$: **15 unknowns $\boldsymbol{\theta}$**

(b) Data

5 species, Days 1–21
4 conditions:
commensal / dysbiotic
× static / HOBIC flow
composition + viability
pH: held out for checking

(c) Staged TMCMC on the GPU

Phase 1

(i) composition initial box → (ii) composition box ±4 SD

Phase 2

(iii) + viability initial box → (iv) + viability box ±4 SD

each stage: tempering $\beta = 0 \rightarrow 1$, resampling, K MH moves (random walk / DE)
implicit solver in JAX, `jit`+`vmap` over all particles on a GPU (∼150×)

accept a stage only if
3 seeds agree
(max log L, medians)
≥ 5 moves / parameter
no mass piled at a bound
else: more particles or wider bound, repeat